

HW 1, Q2(a)

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$$

$$\begin{aligned} \text{take log gives } \log(R_{e,t+1}) &= \log(P_{t+1} + D_{t+1}) - \log(P_t) \\ &= \log(P_{t+1} (1 + \frac{D_{t+1}}{P_{t+1}})) - \log(P_t) \\ &= \log(P_{t+1}) + \log(1 + \frac{D_{t+1}}{P_{t+1}}) - \log(P_t) \end{aligned}$$

$$\text{define } r_{e,t+1} = \log(R_{e,t+1}), \quad p_t = \log P_t, \quad d_t = \log D_t$$

$$\text{then } \frac{D_{t+1}}{P_{t+1}} = \frac{e^{d_{t+1}}}{e^{p_{t+1}}} = e^{d_{t+1} - p_{t+1}}$$

$$\Rightarrow r_{e,t+1} = p_{t+1} + \log(1 + e^{d_{t+1} - p_{t+1}}) - p_t$$

$$\text{now define } f(x) = \log(1 + e^x)$$

use first order Taylor expansion around average value $\bar{x}$,

$$f(x) = f(\bar{x}) + f'(\bar{x})(x - \bar{x}), \text{ where } \bar{x} = d_{t+1} - p_{t+1}$$

$$f'(x) = \frac{e^x}{1+e^x} \quad \text{define } K = \frac{1}{1+e^{\bar{x}}}, \text{ then } f'(\bar{x}) = \frac{e^{\bar{x}}}{1+e^{\bar{x}}} = 1-K$$

$$\Rightarrow f(x) = f(\bar{x}) + (1-K)(x - \bar{x})$$

$$= \underbrace{f(\bar{x}) - (1-K)\bar{x}}_{\text{define as } K_0} + (1-K)x = K_0 + (1-K)x$$

define as K_0

$$\begin{aligned} K_0 &= f(\bar{x}) - (1-K)\bar{x} = \log(1 + e^{\bar{x}}) - (1-K)\bar{x} \quad (e^{\bar{x}} = \frac{1}{K} - 1, \bar{x} = \log(\frac{1}{K} - 1)) \\ &= \log(\frac{1}{K}) - (1-K)\log(\frac{1}{K} - 1) \\ &= -\log K - (1-K)\log(\frac{1}{K} - 1) \end{aligned}$$

$$\text{substitute } x = d_{t+1} - p_{t+1}$$

$$f(x) = K_0 + (1-K)(d_{t+1} - p_{t+1})$$

$$\Rightarrow r_{e,t+1} \approx p_{t+1} + K_0 + (1-K)(d_{t+1} - p_{t+1}) - p_t$$

$$\approx K_0 + (1-K)d_{t+1} + Kp_{t+1} - p_t$$

$$\Rightarrow p_t \approx K_0 + (1-K)d_{t+1} - r_{e,t+1} + Kp_{t+1}$$

subtract d_t on both sides,

$$p_t - d_t \approx K_0 + \Delta d_{t+1} - r_{e,t+1} + K(p_{t+1} - d_{t+1})$$

define $dp_t = d_t - p_t$, then

$$dp_t \approx -K_0 + r_{e,t+1} - \Delta d_{t+1} + K dp_{t+1}$$

$$\approx (-K_0 + r_{e,t+1} - \Delta d_{t+1} + K dp_{t+1})$$

Keep substituting until $t+H$

$$dp_t = -K_0(1 + K + \dots + K^{H-1}) + \sum_{h=1}^H K^{h-1} r_{e,t+h} - \sum_{h=1}^H K^{h-1} \Delta d_{t+h} + K^H dp_{t+H}$$

$$\text{set } S = 1 + K + \dots + K^{H-1}$$

$$KS = K + \dots + K^{H-1} + K^H$$

$$\text{subtraction: } (1-K)S = 1 - K^H \therefore S = \frac{1-K^H}{1-K}$$

$$\Rightarrow dp_t = \frac{K_0(K^H-1)}{1-K} + \sum_{h=1}^H K^{h-1} r_{e,t+h} - \sum_{h=1}^H K^{h-1} \Delta d_{t+h} + K^H dp_{t+H} \quad (1.1)$$

take conditional expectations at time t ,

$$E_t[dp_t] = dp_t$$

$$\Rightarrow dp_t = \frac{K_0(K^H-1)}{1-K} + \sum_{h=1}^H K^{h-1} E_t[r_{e,t+h}] - \sum_{h=1}^H E_t[\Delta d_{t+h}] + K^H E_t[dp_{t+H}] \quad (1.2)$$

let $H \rightarrow \infty$, assume no bubble, $K^H E_t[dp_{t+H}] \rightarrow 0$ as $H \rightarrow \infty$

because $K \in (0,1)$, $K^H \rightarrow 0$ as $H \rightarrow \infty$

$$\Rightarrow dp_t = \frac{-K_0}{1-K} + \sum_{h=1}^H K^{h-1} E_t[r_{e,t+h}] - \sum_{h=1}^H E_t[\Delta d_{t+h}] \quad (1.3)$$

keep substituting until $t+H$

$$dp_t = -k_0 (1 + K + \dots + K^{H-1}) + \sum_{h=2}^H K^{h-1} r_{e,t+h} = \sum_{h=1}^H K^{h-1} \Delta d_{t+h} + K^H dp_{t+H}$$

$$\text{set } S = 1 + K + \dots + K^{H-1}$$

$$KS = K + K^2 + \dots + K^H$$

$$\text{subtraction } (1-K)S = 1 - K^H \quad S = \frac{1-K^H}{1-K}$$

$$\Rightarrow dp_t = \frac{k_0(K^H-1)}{1-K} + \sum_{h=2}^H K^{h-1} r_{e,t+h} - \sum_{h=1}^H K^{h-1} \Delta d_{t+h} + K^H dp_{t+H} \quad (1.1)$$

take conditional expectations at time t ,

$$E_t[dp_t] = dp_t$$

$$\Rightarrow dp_t = \frac{k_0(K^H-1)}{1-K} + \sum_{h=2}^H K^{h-1} E_t[r_{e,t+h}] - \sum_{h=1}^H K^{h-1} E_t[\Delta d_{t+h}] + K^H E_t[dp_{t+H}] \quad (1.2)$$

let $H \rightarrow \infty$, assume no bubble, $K^H E_t[dp_{t+H}] \rightarrow 0$ as $H \rightarrow \infty$

~~because $K \in (0,1)$, $K^H \rightarrow 0$ as $H \rightarrow \infty$~~

$$\Rightarrow dp_t = \frac{-k_0}{1-K} + \sum_{h=2}^H K^{h-1} E_t[r_{e,t+h}] - \sum_{h=1}^H E_t[\Delta d_{t+h}] \quad (1.3)$$

$$Q1(e) \quad \text{because } dy_t = \log\left(1 + \frac{D_t}{p_t}\right)$$

just proved in Q1(a) that

$$r_{e,t+1} = p_{t+1} + dy_{t+1} - p_t$$

subtract Δd_{t+1} on both sides,

$$r_{e,t+1} - \Delta d_{t+1} = p_{t+1} - d_{t+1} + dy_{t+1} + d_t - p_t$$

$$\text{let } dp_t = d_t - p_t = \log \frac{D_t}{p_t} \quad dy_t = \log\left(1 + \frac{D_t}{p_t}\right)$$

$$e^{dy_t} = 1 + \frac{D_t}{p_t}$$

$$e^{dp_t} = e^{dy_t} - 1$$

$$dp_t = \log(e^{dy_t} - 1)$$

$$= \log[e^{dy_t} (1 - e^{-dy_t})]$$

$$= dy_t + \log(1 - e^{-dy_t})$$

$$\Rightarrow r_{e,t+1} - \Delta d_{t+1} = -dy_{t+1} - \log(1 - e^{-dy_{t+1}}) + dy_{t+1}$$

$$+ dy_t + \log(1 - e^{-dy_t})$$

$$= dy_t + \log(1 - e^{-dy_t}) - \log(1 - e^{-dy_{t+1}})$$

(footnote 1)

now define $f(y) = \log(1 - e^{-y})$

use first order expansion at average $\bar{y} = \bar{dy}$

$$f(y) = f(\bar{y}) + f'(\bar{y})(y - \bar{y})$$

$$f'(y) = \frac{e^{-y}}{1 - e^{-y}} \quad \therefore f'(\bar{y}) = \frac{e^{-\bar{dy}}}{1 - e^{-\bar{dy}}}$$

define $K = e^{-\bar{dy}}$, $\therefore f'(\bar{y}) = \frac{K}{1-K}$

$$f(\bar{y}) = \log(1 - e^{-\bar{dy}}) = \log(1 - K)$$

$$f(dy_t) = \log(1 - K) + \frac{K}{1-K} (dy_t - \bar{dy}) \quad (\text{footnote 2})$$

substitute back,

$$\begin{aligned} r_{e,t+1} - \Delta d_{t+1} &= dy_t + \log(1 - K) + \frac{K}{1-K} (dy_t - \bar{dy}) \\ &\quad - \log(1 - K) - \frac{K}{1-K} (dy_{t+1} - \bar{dy}) \\ &= dy_t + \frac{K}{1-K} dy_t - \frac{K}{1-K} dy_{t+1} \\ &= \frac{1}{1-K} dy_t - \frac{K}{1-K} dy_{t+1} \end{aligned}$$

reorganize,

$$dy_t = (1-K)(r_{e,t+1} - \Delta d_{t+1}) + K dy_{t+1}$$

keep substituting until $t+H$

$$\begin{aligned} dy_t &= (1-K) [K^0(r_{e,t+1} - \Delta d_{t+1}) + K(r_{e,t+2} - \Delta d_{t+2}) + \dots] + K^H dy_{t+H} \\ &= (1-K) \left(\sum_{h=1}^H K^{h-1} r_{e,t+h} - \sum_{h=1}^H K^{h-1} \Delta d_{t+h} \right) + K^H dy_{t+H} \quad (1.7) \end{aligned}$$

take conditional expectation at time t , we have (1.8)

$H \rightarrow \infty$, with no bubble assumption, $K^H E_t[dy_{t+H}] \rightarrow 0$

we have (1.9)